

## REVIEW ARTICLE

Edward W. Campion, M.D., *Editor*

## Invasive Candidiasis

Bart Jan Kullberg, M.D., Ph.D., and Maiken C. Arendrup, M.D., Ph.D.

**I**NVASIVE CANDIDIASIS IS THE MOST COMMON FUNGAL DISEASE AMONG HOSPITALIZED patients in the developed world. Invasive candidiasis comprises both candidemia and deep-seated tissue candidiasis. Candidemia is generally viewed as the more common type of the disease, and it accounts for the majority of cases included in clinical trials. Deep-seated candidiasis arises from either hematogenous dissemination or direct inoculation of candida species to a sterile site, such as the peritoneal cavity (Fig. 1). Mortality among patients with invasive candidiasis is as high as 40%, even when patients receive antifungal therapy. In addition, the global shift in favor of nonalbicans candida species is troubling, as is the emerging resistance to antifungal drugs. During the past few years, new insights have substantially changed diagnostic and therapeutic strategies.

From the Department of Medicine and Radboudumc Center for Infectious Diseases, Radboud University Medical Center, Nijmegen, the Netherlands (B.J.K.); and the Department of Microbiology and Infection Control, Statens Serum Institut, Copenhagen (M.C.A.). Address correspondence to Dr. Kullberg at the Department of Medicine, Radboud University Medical Center, P.O. Box 9101, 6500 HB Nijmegen, the Netherlands, or at [bj.kullberg@radboudumc.nl](mailto:bj.kullberg@radboudumc.nl).

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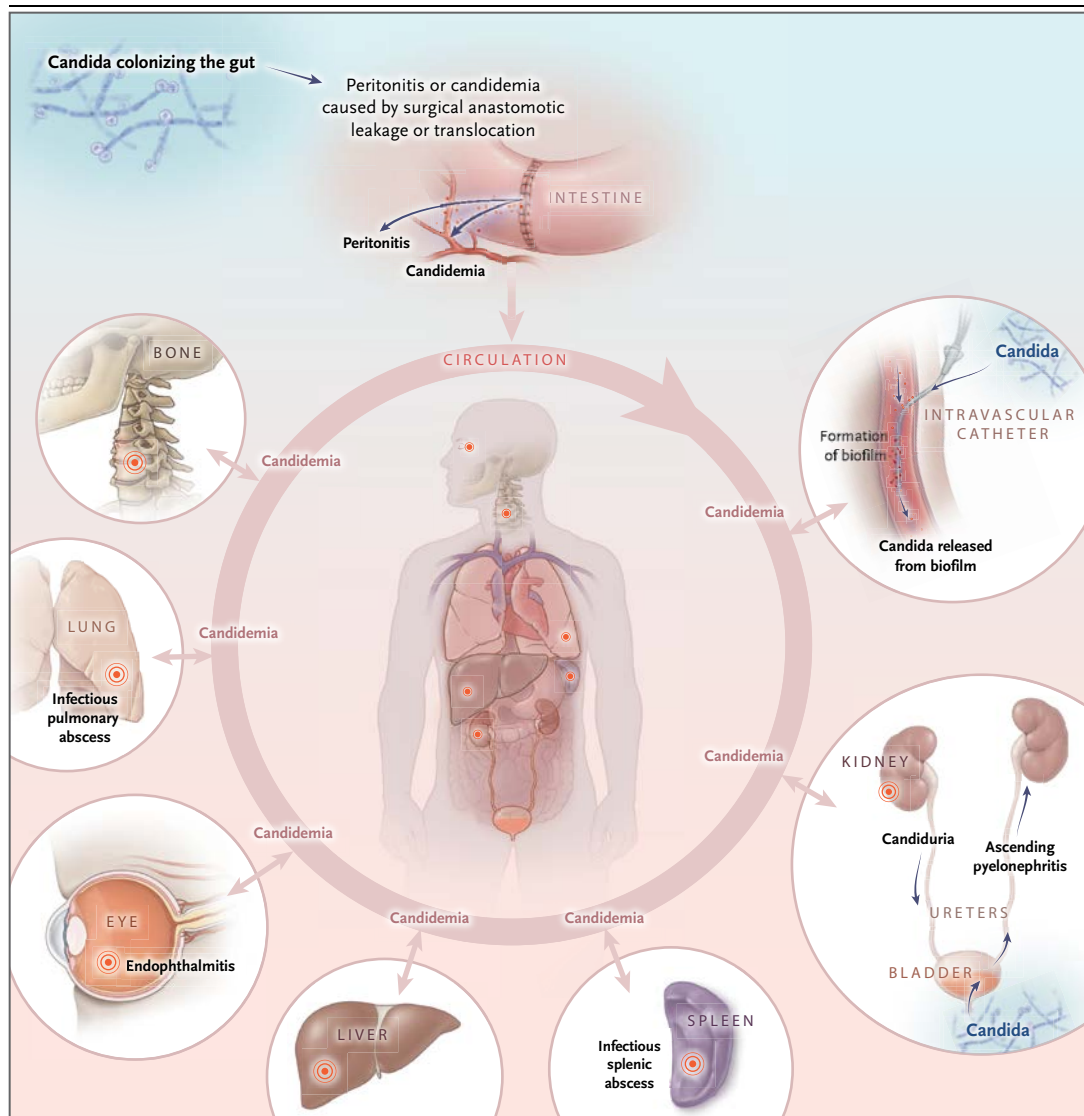
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## EPIDEMIOLOGY

According to conservative estimates, invasive candidiasis affects more than 250,000 people worldwide every year and is the cause of more than 50,000 deaths. Incidence rates of candidemia have been reported to be between 2 and 14 cases per 100,000 persons in population-based studies.<sup>1,2</sup> Candidemia has often been cited as the fourth most common bloodstream infection.<sup>3</sup> Although this statistic applies to intensive care units (ICUs), in most population-based studies candidemia is reported as the seventh to tenth most common bloodstream infection. Incidence rates have been increasing or stable in most regions, although declining rates have been reported in high-incidence areas after improvements in hygiene and disease management were introduced.<sup>2,4,5</sup>

The incidence of candidemia is age-specific, with the maximum rates observed at the extremes of age. Risk factors are summarized in Table 1.<sup>2,6,7</sup> The presence of central vascular catheters, recent surgery (particularly abdominal surgery with anastomotic leakages), and the administration of broad-spectrum antibiotic therapy constitute the major risk factors for invasive candidiasis. Although candidemia has been described as the most common manifestation of invasive candidiasis, blood-culture-negative forms include syndromes such as chronic disseminated (hepatosplenic) candidiasis in patients with hematologic cancers and deep-seated infection of other organs or sites, such as the bones, muscles, joints, eyes, or central nervous system. Infections at most of these sites arise from an earlier or undiagnosed bloodstream infection. Conversely, the direct introduction of candida may occur at a sterile site, resulting, for example, in ascending renal candidiasis or candida peritonitis after intestinal surgery.<sup>8</sup> Deep-seated infections may remain localized or lead to secondary candidemia. The limited published data available suggest that invasive abdominal candidiasis may be much more common than recognized.<sup>8,9</sup>



**Figure 1. Pathogenesis of Invasive Candidiasis.**

Candida species that colonize the gut invade through translocation or through anastomotic leakage after laparotomy and cause either localized, deep-seated infection (e.g., peritonitis), or candidemia. In patients with indwelling intravascular catheters, candidemia that originates from the gut or the skin leads to colonization of the catheter and the formation of biofilm. Fungi are subsequently released from the biofilm, causing persistent candidemia. Once candidemia has developed, whether from a colonized intravascular catheter or by other means, the fungi may disseminate, leading to secondary, metastatic infections in the lung, liver, spleen, kidneys, bone, or eye. These deep-seated infections may remain localized or lead to secondary candidemia. During candidemia, the fungi in the bloodstream may enter the urine, leading to candiduria. Less frequently, deep-seated candidiasis may occur as a result of ascending pyelonephritis and may either remain localized or lead to secondary candidemia.

#### CANDIDA SPECIES

The species distribution has changed over the past decades. Whereas *Candida albicans* had previously been the dominating pathogen, this species today accounts for only half the isolates detected in

many surveys.<sup>1,2,10</sup> *C. glabrata* has emerged as an important pathogen in northern Europe, the United States, and Canada, whereas *C. parapsilosis* is more prominent in southern Europe, Asia, and South America. Changes in species distribution may drive treatment recommendations, given the

**Table 1. Risk Factors for Invasive Candidiasis.\***

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| Critical illness, with particular risk among patients with long-term ICU stay                                       |
| Abdominal surgery, with particular risk among patients who have anastomotic leakage or have had repeat laparotomies |
| Acute necrotizing pancreatitis                                                                                      |
| Hematologic malignant disease                                                                                       |
| Solid-organ transplantation                                                                                         |
| Solid-organ tumors                                                                                                  |
| Neonates, particularly those with low birth weight, and preterm infants                                             |
| Use of broad-spectrum antibiotics                                                                                   |
| Presence of central vascular catheter, total parenteral nutrition                                                   |
| Hemodialysis                                                                                                        |
| Glucocorticoid use or chemotherapy for cancer                                                                       |
| Candida colonization, particularly if multifocal (colonization index >0.5 or corrected colonization index >0.4)     |

\* ICU denotes intensive care unit. For further information see Cleveland et al.,<sup>2</sup> Arendrup et al.,<sup>6</sup> and Lortholary et al.<sup>7</sup>

differences in susceptibility to azoles and echinocandins among these species.

*Candida* species differ considerably in virulence. *C. parapsilosis* and *C. krusei* are less virulent than *C. albicans*, *C. tropicalis*, and *C. glabrata*.<sup>11</sup> This variation is reflected in the low mortality among patients with *C. parapsilosis* candidemia and in the fact that infection with *C. krusei* is highly uncommon except in patients with severe immunodeficiency and prior exposure to an azole.<sup>6</sup> Despite its low virulence, *C. parapsilosis* can thrive in certain clinical settings owing to its ability to adhere to medical devices and its propensity to colonize human skin, characteristics that facilitate nosocomial outbreaks.<sup>12</sup> Other species that appear with less frequency in clinical settings, such as *C. dubliniensis*, *C. lusitanae*, *C. kefyr*, and *C. guilliermondii*, are associated with specific susceptibility patterns or with specific hosts (e.g., *C. dubliniensis* has been particularly common in HIV-infected patients).

#### IMMUNOGENETICS OF CANDIDA INFECTIONS

The majority of patients in the ICU do not acquire invasive candidiasis, even if they share similar risk factors. Thus, it is likely that variation in the genes that confer susceptibility to candida infection makes certain patients more prone to infection. A large clinical study revealed that susceptibility to candidemia was increased among European and North American patients who had single-nucleotide polymorphisms (SNPs) in the

toll-like receptor 1–interferon- $\gamma$  pathway, as compared with a clinical control cohort matched for underlying disease.<sup>13</sup> In a genomewide association study in which susceptibility to candidemia was assessed, three new genes associated with an increased risk of disease were identified. Patients in the ICU who carried two or more alleles at these particular loci had a risk of candidemia that was 19 times as high as the risk among patients who did not have those alleles.<sup>14</sup> Similarly, disease progression and persistent candidemia despite antifungal therapy were associated with cytokine polymorphisms that led to either increased circulating levels of antiinflammatory interleukin-10 or decreased levels of proinflammatory interleukin-12b cytokine.<sup>15</sup> These findings underscore the importance of cytokine balance with respect to both the susceptibility to acquiring invasive candidiasis and the ability to clear the infection once it has been disseminated. The identification of specific alleles related to the risk of disease and of cytokine pathways associated with unfavorable outcomes suggests that screening strategies based on the presence or absence of certain SNPs may help to identify patients at risk who could benefit from prophylactic antifungal treatment or adjunctive immunotherapy.<sup>16</sup>

#### DIAGNOSIS

The armamentarium available for diagnosing invasive candidiasis includes direct detection, in which specimens of blood or tissue from normally

sterile sites are cultured, and indirect detection, in which surrogate markers and polymerase-chain-reaction (PCR) assays are used (Table 2).<sup>18,21,22</sup> No test is perfect, and it is therefore necessary to perform several diagnostic tests to achieve maximal accuracy.

Culture is currently the only diagnostic approach that allows subsequent susceptibility testing. The sensitivity of blood cultures is far from ideal, with a sensitivity of 21 to 71% reported in autopsy studies.<sup>9</sup> Whereas blood cultures may establish a diagnosis during the period when candida resides in the bloodstream, cultures of blood obtained from patients with hematogenous, deep-seated infections often yield negative results because candida has been cleared from the bloodstream at the time the blood sample is collected.<sup>9</sup> Blood cultures are further limited by slow turnaround times and by the fact that a positive result may be revealed only late in the course of disease. Positive blood cultures should prompt the immediate initiation of therapy and a search for metastatic foci.<sup>18,31</sup>

Candida mannan antigens and antimannan antibodies and  $\beta$ -D-glucan are the primary surrogate markers for invasive candidiasis.<sup>18,21,22</sup> The reported performance of assays for these markers varies somewhat according to case mix, the frequency of sampling, and the choice of comparator. Studies that include healthy controls or less severely ill patients may overestimate specificity, since there are many potential sources of contamination of  $\beta$ -D-glucan testing that can produce false positive results, and these are found more frequently in patients at high risk for invasive candidiasis (Table 2). The major diagnostic benefit of  $\beta$ -D-glucan is its negative predictive value for invasive candidiasis in environments in which the prevalence is low to moderate.

A number of in-house PCR tests for the detection of invasive candidiasis have been evaluated. However, limited validation and standardization have hindered their acceptance and implementation.<sup>27</sup> Nguyen et al. reported that an in-house PCR assay had a sensitivity of 89% for deep-seated candidiasis that was not detected on blood cultures.<sup>28</sup> Two commercial PCR tests have been marketed — the SeptiFast and the fully automated multiplex T2Candida Panel, which was released in 2015.<sup>29,30</sup> The T2Candida Panel has recently

been tested in one clinical trial that produced promising results (Table 2).<sup>30</sup>

## PROPHYLAXIS

In view of the high mortality associated with invasive candidiasis, prophylaxis for selected patients in the ICU who are at high risk for the disease would appear to be appropriate. Until now, the use of antifungal prophylaxis in patients in the ICU has received little support from clinical studies, except for its use in specific high-risk groups.<sup>32</sup> In patients who have had recent abdominal surgery and have recurrent gastrointestinal perforations or anastomotic leakage, fluconazole prophylaxis has been shown to be effective.<sup>33</sup> In other selected patient groups in the ICU, the results have been modest at best. Antifungal prophylaxis has generally shown trends toward reducing the incidence of candidemia by approximately 50%, but this strategy has not been shown to improve survival.<sup>34,35</sup> The major challenge is to select individual patients or subgroups that will benefit most from prophylaxis in order to limit the number needed to treat and to avoid the problem of selective pressure that leads to the emergence of resistance.

A recent randomized, placebo-controlled study used targeted caspofungin prophylaxis in patients in the ICU who were determined to be at high risk for invasive candidiasis with the use of a clinical prediction rule.<sup>36</sup> In this study, both serum  $\beta$ -D-glucan levels and cultures were used to define invasive candidiasis. Overall, there were no significant differences between the study groups in the incidence of candidemia, all-cause mortality, the use of antifungal drugs, or the length of stay. In these types of placebo-controlled studies, culture- and biomarker-based end points may be less appropriate, since they are likely to be biased in favor of the group receiving the study drug. At this time, antifungal prophylaxis should be limited to patients in whom it has proved to be beneficial: patients with gastrointestinal anastomotic leakage, patients undergoing transplantation of the pancreas or the small bowel, selected patients undergoing liver transplantation who are at high risk for candidiasis, and extremely low-birth-weight neonates in settings with a high incidence of neonatal candidiasis.

**Table 2. Diagnostic Tests for Invasive Candidiasis.\***

| Test and Specimen Type                                           | Sensitivity                                               | Specificity                                               | Findings from Studies                                                                                                                                                                                                                                                                                                           | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Culture (blood)                                                  | 21–71                                                     | NA                                                        | Per-patient sensitivity (based on autopsy studies) may be underestimated since patients with positive antemortem blood cultures but with no evidence of organ infection on autopsy were not included. <sup>9,17</sup>                                                                                                           | Obtain daily blood cultures (total volume, 40–60 ml in 10-ml bottles for adults) and additional sets during febrile episodes; sensitivity can be increased by including a mycosis bottle. <sup>18</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $\beta$ -D (blood)                                               | 65–100                                                    | 31–79                                                     | Performance depends on cutoff value and no. of positive samples required. <sup>19</sup> Sensitivity is species-dependent: <i>C. krusei</i> , 100%, 3 cases; <i>C. tropicalis</i> , 91%, 11 cases; <i>C. albicans</i> , 83%, 36 cases; <i>C. glabrata</i> , 81%, 26 cases; <i>C. parapsilosis</i> , 72%, 18 cases. <sup>20</sup> | Not specific for candida. Positive test result requires confirmation and identification of infecting organism ( <i>Aspergillus</i> , <i>Pneumocystis jirovecii</i> or candida). <sup>18,21</sup> Many potential sources for contamination: hemodialysis with cellulose membranes, human blood products (immunoglobulins or albumin), amoxicillin–clavulanate or piperacillin–tazobactam, severe bacterial infections, surgical sponges and gauzes containing glucan, and severe mucositis. <sup>22,24</sup> High negative predictive value in several studies with intermediate prevalence. <sup>20</sup> However, limited sensitivity in other studies suggests that negative predictive value may be insufficient in high-risk patients. <sup>19,21,25</sup> Candida mannan antigen and antimannan antibodies tests may be preferable for circumstances in which candida is main fungal pathogen and risk of false positive $\beta$ -D-glucan test is high. <sup>25,26</sup> |
| Candida mannan antigen and anti-mannan antibodies (blood or CSF) | Per patient, 83 (IQR, 79–87); per sample, 62 (IQR, 55–68) | Per patient, 86 (IQR, 82–90); per sample, 96 (IQR, 94–98) | Sensitivity and specificity results were given per patient and per sample. <sup>22</sup> Sensitivity is species-dependent and lower for <i>C. parapsilosis</i> and <i>C. krusei</i> (40–50%) than for <i>C. albicans</i> , <i>C. glabrata</i> and <i>C. tropicalis</i> (80–100%). <sup>26</sup>                                 | Combined antigen–antibody test required for maximum sensitivity. Used to detect blood-culture negative hepatosplenic candidiasis and CNS candidiasis. <sup>22</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| PCR assay (blood)                                                | 82–98                                                     | 87–98                                                     | Patients had candidemia or invasive candidiasis; <sup>27</sup> results based on meta-analysis of range of in-house multiplex PCR assays                                                                                                                                                                                         | PCR formats specific for detection of candida preferred since they are less prone to contamination by airborne fungi and fungal DNA. In general, sensitivities are similar to those of culture results for candidemia and better for deep-seated candidiasis, with shorter turnaround time. Lack of multicenter validation. <sup>27</sup> For deep-seated candidiasis, sensitivity and specificity higher than with $\beta$ -D-glucan. <sup>17,28</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| SeptiFast                                                        | 48–72                                                     | 99                                                        | Results based on meta-analysis <sup>29</sup>                                                                                                                                                                                                                                                                                    | Detects <i>C. albicans</i> , <i>C. glabrata</i> , <i>C. krusei</i> , <i>C. parapsilosis</i> , <i>C. tropicalis</i> , and <i>Aspergillus fumigatus</i> . Labor-intensive. Risk of false positive results for aspergillus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| T2Candida Panel                                                  | 91                                                        | 94                                                        | Multicenter study among 1501 patients (6 of 1501 candidemic) and additional 250 spiked samples <sup>30,†</sup>                                                                                                                                                                                                                  | Detects <i>C. albicans</i> , <i>C. glabrata</i> , <i>C. krusei</i> , <i>C. parapsilosis</i> , and <i>C. tropicalis</i> . Appears promising but validation in higher-risk populations needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

\* CFS denotes cerebrospinal fluid, CNS central nervous system, ICU intensive care unit, IQR interquartile range, NA not available, and PCR polymerase chain reaction. For further information see Cleveland et al.,<sup>2</sup> Arendrup et al.,<sup>6</sup> and Lortholary et al.<sup>7</sup>

† A spiked sample is a negative sample to which candida has been added.

## EARLY TREATMENT

Retrospective observational studies have suggested that early presumptive antifungal therapy (therapy based on symptoms or biomarkers) is associated with reduced mortality among patients with invasive candidiasis.<sup>37</sup> Support has been provided by recent multivariate analyses, which corrected for confounders that were likely to introduce bias in observational cohort studies. These analyses consistently identified the early use of appropriate antifungal therapy and control of the source of infection as major determinants of survival.<sup>38–40</sup> Thus, although it is plausible that early, presumptive treatment of patients with invasive candidiasis is beneficial, such strategies have not been validated by prospective studies.

More refined approaches include treatment that is driven by prediction rules based on clinical risk factors, the presence of candida colonization, and the results of screening for serum  $\beta$ -D-glucan,<sup>25,41</sup> but to date no such approach has been shown to reduce mortality or length of stay in prospective studies. In addition, published prediction rules are not generally applicable in regions or settings that are different from those in the study.<sup>42,43</sup>

The clinical usefulness of prediction rules is affected by the low prevalence of invasive candidiasis.<sup>9,43</sup> In typical ICU settings, where the pretest likelihood of candidiasis is 0.5 to 10%, both individual, non-culture-based tests and risk-factor-based rules, which have a specificity of 50 to 80%, will lead to a positive predictive value of merely 1 to 30%.<sup>42</sup> Rather than being seen as definitive diagnostic tools, prediction rules and nonculture-based tests might be best viewed as markers that aid in the assessment of the possibility that a patient has invasive candidiasis.<sup>9</sup>

## CHOICE OF ANTIFUNGAL THERAPY

Three classes of antifungal drugs are available for the treatment of invasive candidiasis (Table 3), and each new antifungal drug has been compared with a preexisting standard regimen in randomized trials. However, these studies were powered for noninferiority, and prospective studies intended to assess the superiority of one antifungal class of drug over another and to identify the most effective antifungal treatment strategies are unavailable.

Early studies showed that fluconazole, voriconazole, and caspofungin were as effective as amphotericin B deoxycholate and were associated with significantly lower levels of toxic effects and of treatment discontinuation.<sup>44,45,47</sup> The results of such studies marked the end of the use of amphotericin B deoxycholate as a treatment option for invasive candidiasis, except in environments with limited resources.<sup>31</sup> Micafungin was shown to be as effective as caspofungin and liposomal amphotericin B in two subsequent comparative trials.<sup>49,50</sup>

A pivotal study compared the efficacy of anidulafungin with that of fluconazole.<sup>48</sup> Although the study had been designed to assess the non-inferiority of anidulafungin, overall response rates were significantly higher with anidulafungin than with fluconazole (76% vs. 60%;  $P=0.01$ ). The apparent superiority of anidulafungin over fluconazole was most distinct in patients infected with *C. albicans* (global response, 81% vs. 62%;  $P=0.02$ ), even though the *C. albicans* was almost uniformly susceptible to fluconazole.<sup>48</sup> Inferior outcomes with fluconazole were also observed in patients with low scores (indicating less severe disease) on the Acute Physiology and Chronic Health Evaluation (APACHE II), which suggested that inferior outcomes with fluconazole were not related to severity of illness. Post hoc multivariate analyses have not indicated that the differences in outcome with each drug were related to other, confounding factors.<sup>51</sup> Nevertheless, the question of whether a single noninferiority trial can establish the superiority of echinocandins over azoles for the treatment of invasive candidiasis has remained controversial, and opinions among experts in mycology are divided.

More recent studies have provided reasonable support, but no formal proof, for the superiority of echinocandins as treatment for the majority of patients with invasive candidiasis. Most notable is the pooled analysis of patient-level data from seven randomized trials that assessed antifungal treatments.<sup>38</sup> With 30-day all-cause mortality used as an unequivocal end point, the most important finding was that randomization to an echinocandin was associated with better survival rates and greater clinical success than treatment with a triazole or amphotericin B. The improved outcomes were most evident among patients infected with *C. albicans* or *C. glabrata*. The benefit of echinocandin therapy was observed among pa-

**Table 3. Characteristics of Randomized, Controlled Trials for Invasive Candidiasis.**

| Study Regimen                                                  | Comparator Regimen                                                                  | Treatment Duration                                                                                                           | Step-Down Regimen                                                                                                                          | Primary Outcome                                                                         | Standardized Success Rate*                                                        | All-Cause Mortality                                                                | Study                                  |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------|
| Fluconazole, 400 mg/day                                        | Amphotericin B, 0.5–0.6 mg/kg body weight/day                                       | ≥14 Days after last positive blood culture and resolution of clinical signs                                                  | Not allowed                                                                                                                                | Clinical and microbiologic success at last available study visit                        | Fluconazole, 70%; amphotericin B, 79% (P=0.22)                                    | Fluconazole, 40%; amphotericin B, 33% (P=0.20)                                     | Rex et al., 1994 <sup>44</sup>         |
| Caspofungin, 50 mg/day†                                        | Amphotericin B, 0.6–0.7 mg/kg/day (0.7–1.0 mg/kg/day for patients with neutropenia) | ≥14 Days after last positive culture                                                                                         | ≥10 Days, oral fluconazole, 400 mg/day                                                                                                     | Clinical and microbiologic success at end of intravenous therapy                        | Caspofungin, 73%; amphotericin B, 62% (P=0.09)                                    | Caspofungin, 34%; amphotericin B, 30% (P=0.23)                                     | Mora-Duarte et al., 2002 <sup>45</sup> |
| Fluconazole, 800 mg/day, and amphotericin B, 0.6–0.7 mg/kg/day | Fluconazole, 800 mg/day                                                             | Amphotericin B component, 5–8 days; fluconazole, ≥14 days after last positive blood culture and resolution of clinical signs | >5 Days, oral fluconazole, 800 mg/day                                                                                                      | Time to failure (death, alternative therapy, or withdrawal)                             | Fluconazole plus amphotericin B, 69%; fluconazole, 56% (P=0.04)‡                  | Fluconazole plus amphotericin B, 40%; fluconazole, 39% (P=0.89)                    | Rex et al., 2003 <sup>46</sup>         |
| Voriconazole, 3 mg/kg, twice daily†                            | Amphotericin B, 0.7–1.0 mg/kg/day followed by fluconazole, 400 mg/day†              | ≥14 Days after last positive culture                                                                                         | Voriconazole group, >3 days, oral voriconazole, 200 mg twice daily; amphotericin B and fluconazole group: >3 days, fluconazole, 400 mg/day | Clinical and microbiologic success at 12 wk after end of therapy                        | Voriconazole, 65%; amphotericin B and fluconazole, 71% (P=0.25)                   | Voriconazole, 36%; amphotericin B and fluconazole, 42% (P=0.23)                    | Kullberg et al., 2005 <sup>47</sup>    |
| Anidulafungin, 100 mg/day†                                     | Fluconazole, 400 mg/day                                                             | ≥14 Days after last positive culture and improvement of clinical signs                                                       | ≥10 Days, oral fluconazole, 400 mg/day                                                                                                     | Clinical and microbiologic success at end of intravenous therapy                        | Anidulafungin, 76%; fluconazole, 60% (P=0.01)                                     | Anidulafungin, 23%; fluconazole, 31% (P=0.13)                                      | Reboli et al., 2007 <sup>48</sup>      |
| Micafungin, 100 mg/day                                         | Liposomal amphotericin B, 3 mg/kg/day                                               | >14 Days                                                                                                                     | Not allowed                                                                                                                                | Clinical and microbiologic success at end of intravenous therapy, per-protocol subgroup | Micafungin, 74%; liposomal amphotericin B, 70% (P=0.27)                           | Micafungin, 40%; liposomal amphotericin B, 40% (P=0.94)                            | Kuse et al., 2007 <sup>49</sup>        |
| Micafungin, 100 or 150 mg/day                                  | Caspofungin, 50 mg/day†                                                             | ≥14 Days after last positive culture and resolution of clinical signs                                                        | ≥10 Days, oral fluconazole, 400 mg/day                                                                                                     | Clinical and microbiologic success at end of intravenous therapy                        | Micafungin, 100 mg/day, 76%; micafungin 150 mg/day 71%; caspofungin, 72% (P=0.36) | Micafungin, 100 mg/day, 29%; micafungin 150 mg/day, 33%; caspofungin, 26% (P=0.19) | Pappas et al., 2007 <sup>50</sup>      |

\* The standardized success rate was based on the modified intention-to-treat population for the last available study visit<sup>44,46,47</sup> or the end of intravenous therapy.<sup>45,48,50</sup>

† Maintenance doses for fluconazole, caspofungin, voriconazole, and anidulafungin are shown. The loading doses, administered on the first day of treatment, are as follows: fluconazole, 800 mg; caspofungin, 70 mg; voriconazole, 6 mg per kilogram of body weight, two doses; and anidulafungin, 200 mg.

‡ These data are from a secondary analysis. P=0.08 for the primary analysis, which was a Kaplan–Meier time-to-failure analysis.

tients with APACHE II scores in all but the highest quartiles, suggesting that the survival benefit associated with echinocandin treatment is not limited to the sickest patients.<sup>38</sup> In addition to treatment with an echinocandin antifungal agent, the removal of intravenous catheters was an independent determinant of improved survival.<sup>38</sup>

Several cohort studies in which multivariate models were used have consistently identified treatment with an echinocandin and catheter removal as the management strategies associated with better outcomes.<sup>40,52</sup> Additional data have provided reasonable support for the efficacy of echinocandins in patients in the ICU, patients with deep-seated candidiasis, and patients infected with species other than *C. albicans*.<sup>53,54</sup> The observation that success rates among patients infected with *C. parapsilosis* are as good as those among patients infected with other species should be regarded with some caution. *C. parapsilosis* is less susceptible to the echinocandins than other candida species at the cellular and enzyme level and tends to be associated with higher persistence and breakthrough rates among patients receiving an echinocandin.<sup>45</sup>

Clinical trials and hence treatment guidelines are biased toward patients with candidemia, since the infection is easier to recognize and the patients easier to enroll in clinical studies than patients with deep-seated candidiasis. In addition, the comparison of trials is hampered, since the studies have been conducted over an extended period during which many advances in care have been introduced. Despite these caveats, echinocandins are suggested to be associated with better outcomes than those with azoles regardless of the type of invasive candidiasis, APACHE II score, and candida species (except for *C. parapsilosis*), and it is hard to justify withholding these agents as the first choice for treatment.<sup>55</sup> Nevertheless, some experts believe that there is a subgroup of ambulatory, stable, low-risk patients for whom primary therapy with fluconazole is acceptable. Moreover, there are clinical scenarios in which triazoles may be preferred, such as in the treatment of meningitis, endophthalmitis, and urinary tract candidiasis (conditions in which echinocandins are limited by their pharmacokinetics) or in the treatment of patients who have previously been exposed to echinocandins for prolonged periods.

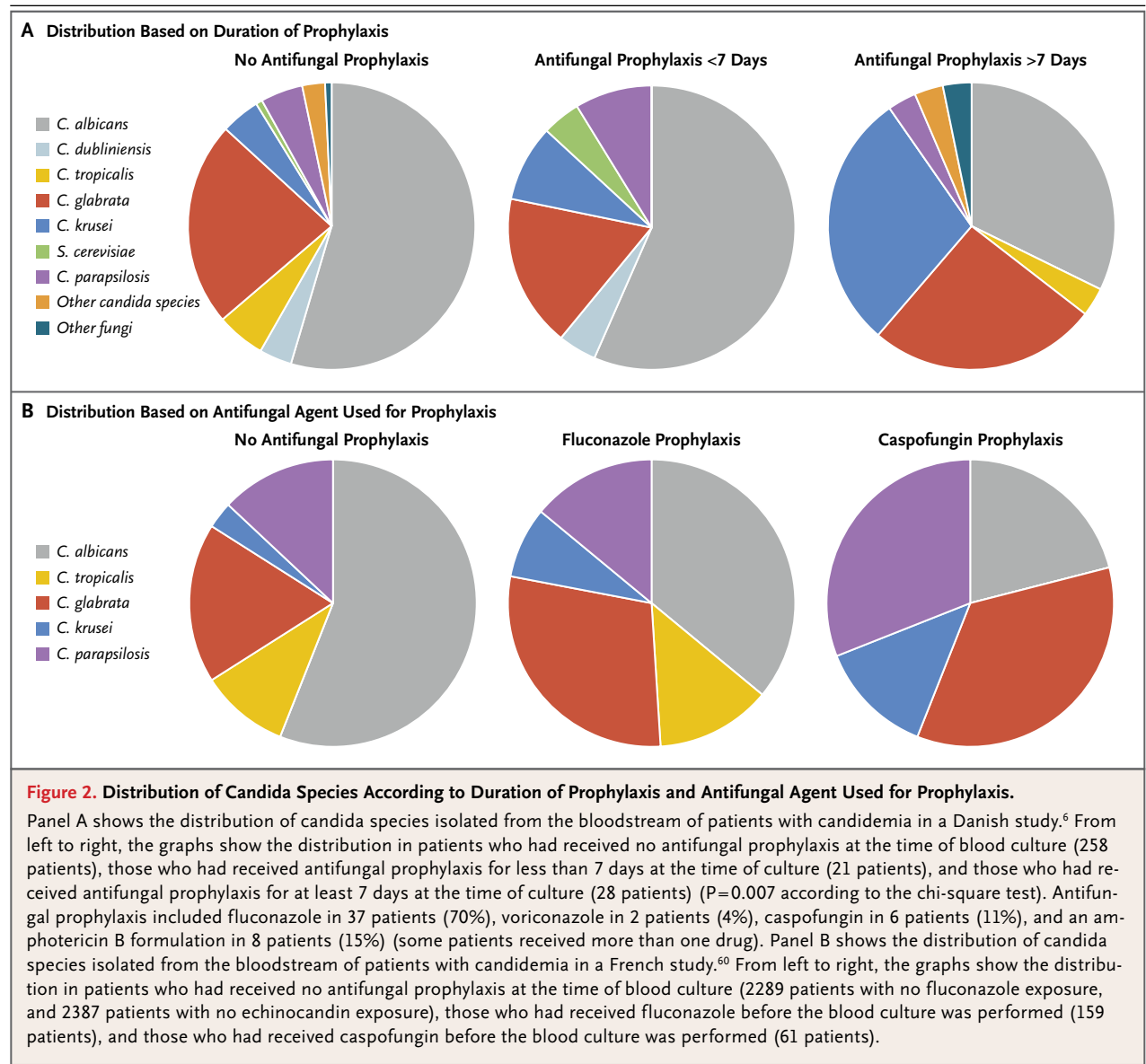
#### DURATION OF THERAPY AND STEP-DOWN CARE

Few data are available to support recommendations regarding the total duration of therapy or the step-down procedure from echinocandins to intravenous or oral azoles.<sup>56</sup> Since it is assumed that initial therapy with echinocandins is most effective in preventing death, the primary requirement for the transition to azoles should be the clinical stabilization of the patient rather than identification of the infecting species and its susceptibility to azoles only — with the probable exception of *C. parapsilosis* infection.

Recent phase 4 studies have incorporated a step-down strategy to an oral azole as early as 5 days after the start of intravenous treatment with an echinocandin, provided that the infecting candida species has been cleared from the bloodstream and is probably susceptible to azoles and that the patient's condition is clinically stable and the patient is capable of taking oral therapy.<sup>54</sup> The outcomes of a strategy of early step-down with respect to efficacy and survival were similar to those reported in previous studies in which a minimum of 10 days of parenteral echinocandin therapy were required.<sup>54</sup> However, the intent of these studies was not to compare the effects of early step-down therapy with prolonged echinocandin therapy in a randomized fashion, and the patients who underwent the transition to azoles were less severely ill than other patients.

#### CATHETER MANAGEMENT

The concept supporting removal of intravascular catheters in patients with candidemia is based on the identification of catheters as a major risk factor for candidemia, the presence of biofilms of candida species attached to catheters, and the observation that candidemia may persist until catheters have been removed. However, no blinded, randomized studies have been designed to determine the effect of catheter removal on outcomes and mortality. It is unlikely that such a study will ever be performed, and retrospective subgroup analyses have shown divergent outcomes.<sup>38,57,58</sup> Although a careful analysis could not identify a significant effect of early catheter removal within 24 or 48 hours after initiation of treatment,<sup>57</sup> other studies found that catheter



removal at any time point was associated with a reduction in mortality and higher clinical success rates.<sup>39,40,58</sup> In the pooled patient-level analysis of seven randomized treatment trials, treatment with an echinocandin and catheter removal were identified as the two modifiable management strategies associated with better survival.<sup>38</sup> Because patients have to be alive to have a catheter removed, these types of analyses may not be free of bias. Although the debate about this issue will continue, it seems wise to remove all intravascular catheters in patients with candidemia, if logistically feasible.<sup>31,55,59</sup>

#### EMERGING RESISTANCE

Resistance to antifungal treatment can emerge either by means of the selection of species with intrinsic resistance or an induction of resistance in isolates from species that are normally susceptible. The former route is the most common, as illustrated by the emergence of *C. glabrata* after the introduction of fluconazole and of *C. parapsilosis* in settings in which there was increased use of echinocandins (Fig. 2).<sup>6,60</sup> In addition, insufficient dosing of azoles has been associated with the emergence of resistance.<sup>61</sup>

*Candida* isolates with acquired resistance to echinocandins have been reported with increasing frequency.<sup>62</sup> *C. glabrata* is overrepresented among echinocandin-resistant isolates, with resistance rates of 2 to 5% and up to 8 to 12% at selected centers for tertiary care.<sup>62,63</sup> Acquired resistance to echinocandins has also been reported for *C. albicans*, *C. tropicalis*, *C. krusei*, *C. kefyr*, *C. lusitanae*, and *C. dubliniensis*.<sup>62</sup> Recent studies indicate that the rate of acquired resistance to echinocandins in isolates from sources other than blood may be underestimated, which suggests that deep-seated candidiasis may act as a hidden reservoir of echinocandin resistance.<sup>64</sup>

### CONCLUSIONS AND FUTURE PERSPECTIVES

The management of invasive candidiasis has changed markedly during the past decade. Changes in epidemiology and the emergence of resistance, against both triazoles and echinocandins, merit vigilance. We have entered a new era in which better outcomes for patients with invasive candidiasis are less likely to result from new drugs than from early intervention strategies that are based on a combination of clinical prediction rules, non-culture-based tests (e.g., PCR assays or tests for antigens), and, ultimately, personalized,

immunogenetics-based risk profiles. At present, the most important need is for studies that will validate the role of nonculture-based diagnostics in presumptive early treatment strategies.

Accumulating evidence points to the importance of early and appropriate antifungal treatment as a major driver of outcomes. Echinocandins appear to be the drugs of first choice for most patients, irrespective of the severity of illness. This development has marked a paradigm shift in the treatment of invasive candidiasis: treat early with an echinocandin and step down early to a triazole, giving consideration to the clinical stabilization of the patient, the *Candida* species, and its susceptibility. By defining the most effective approach to the management of invasive candidiasis, we may finally begin to see declining mortality among patients with candidemia.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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